

Dr. Vishweshwar Peddy

Senior Director

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EXECUTIVE LEADERSHIP PROFILE

Pharmaceutical development R&D leader with ~20 years of pharmaceutical industry experience following post-doctoral research in the USA, spanning generic API development, CDMO operations, innovator drug discovery (NCEs), with deep expertise in polymorphism, crystallization, particle engineering, and solid-state materials characterization.

Recognized internationally as a leading authority in solid-form science (polymorphism and pharmaceutical cocrystals) with a proven track record of inventing IP-rich solid forms, developing crystallization processes with PAT tools, particle engineering, resolving high-risk development and scale-up challenges, and enabling successful regulatory filings and commercialization.

Demonstrated strength in leadership, including strategy definition, client governance, CAPEX planning, infrastructure build-out, people development, cross-functional integration, while consistently delivering robust, scalable, and defensible development solutions.

CORE LEADERSHIP & TECHNICAL DOMAINS

- Polymorphism, Salts, Cocrystals, Amorphous & ASDs of APIs
- Crystallization process development, Technology transfer & Scale-up
- Modelling simulations for Polymorph predictions
- Particle Science & Engineering
- PAT-Enabled development (FBRM, PVM and ReactIR)
- Filtration & Downstream Processing
- Novel Solid-forms of APIs inventions for DMF filings for Para-IV ANDAs
- Intellectual property & Non-infringement evaluation of novel solid forms
- Regulatory evaluation & DMF Deficiencies of Solid forms/Polymorphism
- CAPEX, Infrastructure & Capability Development
- People Leadership, Mentoring & Talent Development

PROFESSIONAL EXPERIENCE

Sai Life Sciences Ltd., Hyderabad

Senior Director & Head – Particle Science & Engineering (PS&E)

2023 – Present

- Head of Particle Science & Engineering, with end-to-end accountability for materials science, solid-state form selection, crystallization process development, PAT integration, and particle engineering across global innovator programs.
- Defined and institutionalized solid-form and crystallization strategies to consistently ensure:
 - Systematic version and solid-form screening of NCEs/APIs and selection for development.
 - Robust crystallization process development to achieve solid-phase-pure polymorph and/or targeted PSD specifications (D50/D90) of APIs.
 - Plant-scalable processes with targeted yield, filtration performance, and product quality.
- Institutionalized PAT-enabled crystallization process development, integrating FBRM, PVM, ReactIR, and solubility/MSZW data (Crystal16) to enable data-driven design space, process robustness, and confident scale-up.
- Led the resolution of high-risk crystallization and solids-handling challenges, including oiling-out, scaling, slow filtration, and low yield - often under aggressive, client-critical timelines.
- In both FFS and FTE engagement models, delivered 40+ client programs spanning early and late-phase APIs.
- Provided scientific leadership for execution of lab scale-down models, technology transfer, and support of plant campaigns.
- For Materials science and Crystallization process development, serving as the primary technical expert for global clients, leading program reviews, scientific strategy alignment, and complex troubleshooting discussions.
- Led the preparation of FFS and FTE proposals in response to client RFPs and, upon project award, drove end-to-end execution through structured weekly client updates, resulting in consistent year-on-year revenue growth.
- Championed capability building and technical excellence, delivering structured internal training programs (Sai Academy) on polymorphism, crystallization, and PAT for PR&D, PE, and AR&D teams.
- Ensured sustained regulatory compliance, operating within GLP/cGMP/GDP environments and providing scientific support during client audits.
- Led CAPEX planning and secured the procurement of advanced technologies and equipment aligned with client requirements.

Glenmark Life Sciences Ltd. (API Division) (Rebranded as Alivus Life Sciences Ltd.)

Senior Principal Scientist – Process Chemistry, Crystallization & Polymorphism

2018 – 2023

- Provided end-to-end leadership across crystallization, materials science, and process chemistry, supporting route selection, cost-effective API development across all stages, technology transfer, and plant validation. Ensured scientific alignment between R&D and manufacturing to enable smooth scale-up and reliable execution.
- Led and mentored process chemistry teams in the identification, evaluation, and development of scalable and economical routes of synthesis (RoS), including API cost-improvement initiatives. Played a key role in supporting plant validation campaigns, resolved scale-dependent issues, and enabling uninterrupted commercial supply & US-DMF filings.
- Invented novel routes of synthesis, solid forms, and crystallization processes, resulting in multiple patent filings and contributions to several US-DMF submissions, thereby strengthening product differentiation, IP protection, and long-term business competitiveness.
- Served as the organizational polymorph and crystallization technical expert for all the API programs, provided strategic guidance and hands-on problem resolution for polymorphism-related risks in both development and commercial APIs, including phase transformation, form stability, and solid-state characterization and control strategies etc.

Dr. Reddy's Laboratories Ltd.

Principal Scientist

2006 – 2018

- Served as a polymorph scientific expert leader at Integrated Product Development (Generic API), and Drug Discovery Research (NCEs).
- Led Polymorph screening and analytical teams (within the polymorph department).
- Defined and institutionalized solid-form and crystallization strategies:
 - Conducted systematic solid-form screening (including polymorphs and co-crystals) of APIs, resulting in invention of multiple novel solid forms.
 - Characterized novel solid forms using a suite of orthogonal analytical techniques and evaluated their novelty against solid forms reported in patents and literature. Novelty was further validated through external expert review for formal assessment reports.
 - Identified novel crystallization processes for various novel and known solid forms of APIs.
 - Secured patent protection for all novel solid forms and their crystallization processes.

- Obtained regulatory approval, as required, for development programs such as Lenvatinib Mesylate•MIBK solvate. Subsequently developed a robust crystallization process which led to a US-DMF — representing a first-of-its-kind achievement globally.
 - Invention of novel solid forms led to 8 US-DMF filings.
- Leveraged advanced molecular modelling platforms (e.g., Materials Studio and COSMOtherm softwares) to predict solid-form landscapes by evaluating intermolecular interactions, lattice energies, solvation effects, and thermodynamic stability.
- Pioneered high-throughput solid-form screening – evaluated, customized the entire platform as per generic requirements, ensured successful installation and utilization at IPDO, Dr Reddy's, first-time in India
 - Fully automated robotic HTS platform
 - High-throughput PXRD for large-scale solid-form characterization
- Identified and developed multiple crystallization processes for IP-free/Open solid forms, enabling IP positioning and protection through process innovation.
- Established and led an analytical team to conduct stress and indicative ICH stability studies on various novel and IP-free/Open solid forms. Based on stress stability data, identified and recommended appropriate packaging options for the target solid form.
- Collaborated with portfolio and IP teams to select viable solid forms (novel or IP-free/Open forms) of APIs for different geographies, based on crystallization process feasibility, patent landscape evaluation, and stability data.
- Recognized as internal polymorph expert, supported IP, legal, and regulatory assessments.
- Recipient of multiple innovation and excellence awards.
- Authored multiple peer-reviewed research publications on polymorphism and crystallization during tenure at Dr. Reddy's, contributing to advancements in solid-state understanding and crystallization science.

SCIENTIFIC & TECHNICAL LEADERSHIP

Solid-Form (Polymorph) & Crystallization Science

- Led end-to-end solid-form strategy, including version selection (salts, cocrystals) for NCEs and APIs, followed by systematic polymorph screening to identify thermodynamically stable forms for NCEs and novel solid forms for generic API development.
- Developed and optimized robust crystallization processes to consistently deliver solid-phase-pure polymorph, yield, and product quality at laboratory scale, followed by successful and reproducible scale-up at plant.
- Skilled in achieving target Particle Size Distribution (PSD) through both controlled crystallization pathways and downstream particle-size-reduction technologies

such as jet milling and wet milling, enabling consistent manufacturability and formulation compatibility. After successful lab batches, transferred the technology to the plant and successfully scaled up.

- Institutionalized PAT-driven development frameworks, integrating FBRM, PVM, ReactIR, and solubility/MSZW profiling (Crystal16) to enable data-driven design space, mechanistic understanding, and high-confidence scale-up.
- Provided deep technical leadership in troubleshooting complex crystallization and solids-handling challenges, including oiling-out, fouling/scaling, phase instability, inconsistent purity, and purification bottlenecks, ensuring robust resolution and smooth plant execution.

Modelling & In-Silico Tools

- Applied advanced Crystal Structure Prediction (CSP) methodologies and polymorph risk assessment frameworks to proactively identify thermodynamically stable and metastable solid forms, enabling informed solid-form selection and development-risk mitigation.
- Leveraged molecular and crystal engineering tools, including Materials Studio for polymorph prediction and CosmoTherm for solvate screening and rational pharmaceutical cocrystal design, to guide experimental strategy and accelerate solid-form discovery.
- Authored a peer-reviewed publication in *Angewandte Chemie* on polymorph prediction, demonstrating recognized scientific leadership in computational solid-state chemistry and predictive crystallization science.
- Utilized process modelling tools such as Dynochem and MixIT to support crystallization process understanding, mixing and heat-transfer evaluation, and successful scale-up from laboratory to plant, improving process robustness and scalability.

Global Authority in Pharmaceutical Cocrystals

- Subject Matter Expert (SME) in pharmaceutical co-crystals with global recognition, having invented and characterized numerous API co-crystals using Crystal Engineering approaches and Cambridge Structural Database (CSD)-driven design. Highly experienced in advanced solid-state characterization and regulatory differentiation between co-crystals and salts.
- At Dr. Reddy's, invented the Roxadustat–L-Proline co-crystal, filed the patent, and secured regulatory approval leading to US DMF filing (#35463)—the first API co-crystal DMF from India.
- Filed several patents on pharmaceutical cocrystals.
- Authored one book chapter, one highly cited review (>1,250 citations), and 15+ research articles on pharmaceutical co-crystals.

Intellectual Property & Regulatory

- 79 total patent filings
 - 14 granted patents (12 in USA, 1 in Australia and 1 in India)
 - 30 PCT/US published applications
 - 36 Indian complete/provisional filings
- Led novelty assessment of >200 solid forms via literature and patent landscape analysis.
- Conducted non-infringement evaluations for >15 solid forms in collaboration with IP teams and patent attorneys.
- Guided patent claim drafting and strengthened protection of solid forms, crystallization processes, and Route of Synthesis (RoS).
- Led DMF deficiency resolutions related to polymorphism.
- At Dr Reddy's served as polymorph expert for novelty evaluation of polymorphs of APIs.

Analytical, Infrastructure & Collaboration

- 12+ years of leadership in solid-state analytical R&D, with end-to-end responsibility for procurement, installation, qualification, and operation of advanced analytical platforms, including PXRD, HT-PXRD, DSC, TGA, FT-IR, HPLC, GC, Karl Fischer titration, and dissolution systems.
- Provided analytical and stability-science leadership for the evaluation of ICH stability of 200+ solid forms, encompassing crystalline (anhydrous, hydrates, solvates), amorphous systems, amorphous solid dispersions (ASDs), and pharmaceutical cocrystals.
- Recognized expert in single-crystal growth and crystallography, having solved and refined 300+ crystal structures, supporting structural and solid-form elucidation and polymorph purity assessment.
- Led high-potency laboratory upgrades and containment qualification, enabling safe handling of compounds with OEL limits $\leq 0.5 \mu\text{g}/\text{m}^3$, and supporting development of oncology and high-potency APIs.
- Drove digitalization and data integrity initiatives, including implementation of electronic laboratory notebook (e-LNB) systems, improved data traceability, and sustained GLP compliance.
- Served as a trained Safety Auditor through DuPont safety programs, contributing to proactive risk assessment, employee and laboratory safety governance, and good EHS practices.
- Initiated and led strategic academic and scientific collaborations under Confidential Disclosure Agreements (CDAs) with leading national and international institutions—including MIT (USA), Stony Brook University (USA), IIT Madras, IIT Hyderabad, University of Hyderabad, IISER Kolkata, and IICT Hyderabad—to leverage external expertise and advance innovation in materials science.

People, Culture & Execution Leadership

- ~20 years of people leadership, managing teams consisting of scientists and engineers.
- Interviewed and hired 60+ scientists across R&D functions.
- Led full performance management cycles (KRAs, reviews, ratings, succession).
- Mentored PhD and MS students during Post-Doctoral studies
- Implemented Lean Daily Management (LDM) and Kaizen initiatives.
- Extensive project execution experience using Concerto project-management systems

ACADEMIC BACKGROUND

Post-Doctoral Research – University of South Florida, USA (2004-2006)

(Industry-sponsored pharmaceutical development projects in collaboration with Transform Pharmaceuticals and Johnson & Johnson, focused on API development)

PhD – University of Hyderabad, India (1999-2004)

Specialization: Polymorphism, Crystallization, Crystal Engineering, Modelling/Simulations, X-ray Crystallography, Small molecules synthesis, Solid state characterization

SCIENTIFIC IMPACT & RECOGNITION

- Recognized scientific leader with a significant global research footprint, evidenced by 40 peer-reviewed publications, >6,700 citations, and an H-index of 24, demonstrating sustained, high-impact contributions to pharmaceutical solid-state science and crystallization.
- Trusted peer reviewer for leading international journals, including ACS and RSC publications, contributing to scientific quality, rigor, and advancement in crystallization, polymorphism, and solid-state research.
- Invited speaker at prestigious international conferences and forums, including IUCr-PPXRD, two Indo-US bilateral meetings, and international scientific workshops, delivering expert insights on polymorphism, crystallization science crystallography and pharmaceutical cocrystals.
- Recipient of multiple innovation and excellence awards at Dr. Reddy's Laboratories, recognizing sustained leadership in complex product development, scientific innovation, idea generation, and cross-functional execution.